Regularity holds in ZF minus Foundation (via the cumulative hierarchy)

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the closing sentence notes ZFminus thereby proves Foundation => Regularity, completed by the converse in the next proposition; the proof's final citation line has no period in the source - kept as is.

Editorial note. The label zfentailsregularity names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Every set sits inside a transitive closure. Same unit, \ollabel{subsetoftrcl} (source line 466). Line 510: A subset trcl(A) and trcl(A) is transitive.
- Transitive sets are bounded in the hierarchy. Same unit, \ollabel{lem:TransitiveWellFounded} (source line 477). Line 511: the transitive set trcl(A) is covered by some V_alpha.

Source reference. Original lemma lem-TransitiveWellFounded: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 476–476. Prerequisite pointer invoked at proof line(s) 510; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference subsetoftrcl: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 466–466. Prerequisite pointer invoked at proof line(s) 510; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *Regularity holds.*

Proof. Fix A; now $A \subseteq \operatorname{trcl} A$ by subsetoftrcl, which is transitive. So there is some α such that $A \subseteq \operatorname{trcl} A \subseteq V_\alpha$ by lem-TransitiveWellFounded □